Shikhar Shiromani

AI Safety and Interpretability Researcher

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Summary

- AI safety and interpretability researcher working on chain-of-thought faithfulness, steganographic reasoning, and the monitorability of large language models. Machine Learning Engineer at Apple; Pivotal Research Fellow and MARS V Research Fellow.

Research Fellowships

Pivotal Research Fellow — Feb 2026 – present

Chain-of-thought faithfulness in LLMs, Pivotal Research (London)
- Selected as a Research Fellow from 2,500+ applicants; awarded a merit-based extension in March 2026.
- Working with Noah Y. Siegel (Google DeepMind, UK) on chain-of-thought faithfulness.
- Showed CoT monitors collapse from ~95% to under 11% catch rate on the subset where the reasoning trace is the only signal, and that the attack transfers across monitor families and agent models.

MARS V Research Fellow — July 2026 – present

Steganographic reasoning in LLMs, Meridian (Cambridge, UK)
- Selected as one of 86 fellows from 1,733 applications (approximately 5%).
- Working with Julian Schulz on how LLMs hide chain-of-thought in text.
- Evidence for a distributed mechanism that transfers across encoding schemes by reusing the same weights.

SPAR Fellow — Feb 2026 – May 2026

Attention Consistency Training for Safety, Kairos
- Accepted from a round of nearly 2,500 applications.

Research

A False Average - CoT Monitor Collapse — May 2026 – Aug 2026

arXiv 2608.00583
- Gradient-free rewrite of an agent's reasoning, holding every command and output fixed, drops a held-out monitor's catch rate from ~95% to under 11% on CoT-only hacks.
- Demonstrated that aggregate monitor accuracy is a false average that hides near-total collapse on the subset monitors exist to cover.
- Trace-only defenses recover it only partially; a probe on an open-weight surrogate monitor's activations separates missed hacks, with a causal control showing this is a detector rather than latent monitor knowledge.

The Hypocrisy Gap - Internal Belief vs. CoT via Sparse Autoencoders — Nov 2025 – Jan 2026

arXiv 2602.02496
- Introduced a mechanistic metric quantifying divergence between a model's internal belief and its generated explanation using sparse autoencoders and sparse linear probes.
- AUROC 0.55-0.74 for detecting sycophantic and hypocritical runs on Gemma, Llama, and Qwen, consistently outperforming a decision-aligned log-probability baseline (0.41-0.50).

Benchmarking Pluralistic Alignment (PAB) Mar 2026 – June 2026
ICML 2026 Pluralistic Alignment Workshop
- Built the Pluralistic Alignment Benchmark covering 320 scenarios, 8 behavioral-risk categories, and 63 synthetic personas.
- Evaluated 9 leading LLMs with a multi-judge pipeline, revealing substantial persona-conditioned drift in manipulation, concealment, persuasion, and sycophancy.

VLM Reliability Probe Dec 2025 – June 2026
ICLR 2026 Multimodal Intelligence and Multimodal Reasoning Workshops
- Cross-family reliability probe testing whether spatial attention structure predicts VLM correctness.
- Attention coherence has near-zero correlation with accuracy (R approx 0.001); self-consistency is far stronger (R = 0.43).
- Causal ablations expose a split between late-fusion and early-fusion architectures with direct monitor-design implications.

COMPASS - Context-Modulated PID Attention Steering Aug 2025 – Nov 2025
AAAI 2026 LaMAS Workshop (Oral), arXiv 2511.14776
- Interpretable feedback-driven decoding framework steering attention via a Context Reliance Score.
- Reduced contextual hallucinations by 2.8-5.8 points across HotpotQA, XSum, HaluEval, and RAGTruth without retraining.

Linear Predictability of Attention Heads Aug 2024 – May 2025
arXiv 2603.13314
- Revealed emergent inter-head linear structure enabling cross-head QKV approximation from 2-5 reference heads.
- Showed the structure is learned rather than architectural, emerging during pretraining, supported by a theoretical lower bound at initialization.
- Compressed the KV cache via learned projections for 2x reduction with minimal performance loss.

ChameleonBench - Quantifying Alignment Faking May 2025 – June 2025
ACML 2025 (Oral)
- 800-prompt benchmark across 8 misbehavior categories with an external judge pipeline.
- Across 6 leading LLMs, up to 20% higher harmful response rates in free-form versus test-like settings.

ProMoral-Bench - Prompting Strategies for Moral Reasoning and Safety May 2025 – Nov 2025
NeurIPS 2025 PersonaNLP, arXiv 2602.13274
- Benchmark comparing 11 prompting strategies across 4 LLM families, including the new ETHICS-Contrast dataset.
- Introduced the Unified Moral Safety Score; exemplars outperform multi-hop reasoning on accuracy, safety, and efficiency.

SAGE - Agentic Framework for Pathology Biomarker Discovery July 2025 – present
arXiv 2602.00953
- Agentic framework discovering pathology biomarkers grounded in biological literature and multimodal evidence.
- Links image-derived features with molecular markers and clinical outcomes to improve clinical translatability.

Experience

Apple, Machine Learning Engineer — San Diego, CA
June 2026 – present
4 months

- Developing intelligent systems that optimize power and performance across Apple's operating systems.

NVIDIA, Senior Software Engineer, Tech Lead — Santa Clara, CA
Feb 2025 – June 2026
1 year 5 months

- Built the first server lifecycle management system at NVIDIA, streamlining resource requests, allocations, and deployment tracking for faster data center rack bring-up.
- Explored ML techniques to enhance node uptime by predicting failures and optimizing system reliability.

NVIDIA, Machine Learning Research Intern — Santa Clara, CA
May 2024 – Aug 2024
4 months

- Architected an ML-driven predictive maintenance system for 1000+ Blackwell and Hopper nodes, delivering over $100k/year in operational savings.
- Implemented multidimensional multimodal anomaly detection with negative sampling and integrated gradients for variable attribution during model interpretation.
- Built microservices for telemetry collection (6200+ metrics in 2 minutes), Grafana dashboarding, and failure prediction.
- Deployed an Airflow MLOps pipeline and a FastAPI REST service handling 10,000+ inferences/day.
- Presented a poster at NTech Conference 2024. Filed for patent as first author.

AI Institute for Advances in Optimization (AI4OPT), Georgia Tech, Graduate Research Assistant, advised by Prof. Pascal Van Hentenryck — Atlanta, GA
Aug 2023 – Dec 2024
1 year 5 months

- Founding backend engineer for a highly available on-demand multi-modal transportation system focused on large-scale ride-sharing, shipped on iOS and Android.
- Piloted in Savannah, GA over 15 months, scheduling 100+ rides daily.
- Project awarded one of the SMART 20 Awards 2025.

NVIDIA, Software Engineer II — Bangalore, India
July 2019 – July 2023
4 years 1 month

- Architected a scalable self-serve CI/CD platform, cutting engineering workload turnaround time by 93%.
- Presented a paper at NTech India Conference on the benefits of the system.
- Delivered the NvCCP (NVIDIA Camera Control Protocol) module for automotive safety assessments, driving software architecture, features, and functional verification.

Education

Georgia Institute of Technology, Computational Science and Engineering — Atlanta, GA, USA
Aug 2023 – Dec 2024

Birla Institute of Technology and Science, Pilani, Electrical and Electronics Engineering — Pilani, India
Aug 2015 – June 2019

Service and Mentorship

Peer Reviewer — Sept 2025 – present

12 completed reviews recorded on OpenReview

- Efficient Reasoning Workshop, NeurIPS 2025 (3 reviews) and COLM 2026 (5 reviews, repeat invitation from the same organizing committee).
- BlackboxNLP 2026 at EMNLP 2026, Budapest (4 reviews), nominated by the program chair committee.
- Reviewer pool, InterpScience 2026 at NeurIPS 2026; reviewer invitation, Pluralistic Alignment Workshop at ICML 2026.

Research Mentor, Algoverse July 2025 – present

ML Researcher and Summer 2026 mentor

- Invited back as a research mentor for the Summer 2026 cohort.

IEEE Senior Member June 2026 – June 2026

Elevated June 2026

- Highest IEEE grade available by application, held by approximately 10% of members.

Patent Pending (first author) Mar 2025 – Mar 2025

Machine Learning-Based Anomaly Detection Using Data Center Metrics

- Filed March 2025, developed at NVIDIA against Blackwell and Hopper node telemetry.

Technical Skills

Languages: Python, C/C++, TypeScript, JavaScript, SQL

ML and Interpretability: PyTorch, sparse autoencoders, linear probes, activation patching, integrated gradients, TransformerLens-style analysis

Systems and Infrastructure: Kubernetes, Docker, Airflow, FastAPI, Django/Flask, NodeJS, ReactJS, MySQL, InfluxDB, Jenkins

Awards and Honors

Pivotal Research Fellowship: Selected from 2,500+ applicants, plus a merit-based extension (2026)

MARS V Fellowship: 86 selected from 1,733 applications, approximately 5% (2026)

SPAR Fellowship: Kairos, from a round of nearly 2,500 applications (2026)

IEEE Senior Member: Elevated June 2026

SMART 20 Awards 2025: AI4OPT on-demand multi-modal transportation project

Competitively selected orals: ACML 2025 (ChameleonBench), LaMAS at AAAI 2026 (COMPASS), SELVA at ACL 2026 (Linear Predictability, one of four orals)